

A census of O-type stars in the Small Magellanic Cloud: physical properties, masses, ages, and spatial distributions

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ABSTRACT

A comprehensive analysis of the known O-type star population in the Small Magellanic Cloud is presented, based on a sample of 448 spectroscopically classified stars. Photometric data were compiled from multiple catalogues to construct a uniform dataset. Stellar parameters, including effective temperatures and luminosities were derived, and stellar masses and ages were estimated using the BONNSAI Bayesian inference tool. The resulting mass distributions show strong agreement between evolutionary models, while the inferred ages exhibit larger systematic differences between the model grids, but overall indicate a predominantly young stellar population with ages extending to just over ~ 10 Myr. The derived initial mass function is found to be steeper than the canonical Salpeter slope at the high-mass end, likely reflecting a combination of statistical limitations and observational biases. The spatial distribution of O-type stars shows a strong correlation with regions of active star formation, with a small number of isolated objects that may represent runaway systems. The total hydrogen ionising photon output is dominated by O-type stars, with a significant additional contribution from Wolf–Rayet stars, yielding a combined output of 5.38×10^{51} photons s^{-1} . Overall, these results provide a consistent picture of a young, spatially distributed massive stellar population in the Small Magellanic Cloud.

Key words: stars: massive – stars: early-type – Magellanic Clouds

1 INTRODUCTION

High-mass stars play a fundamental role in shaping the structure and evolution of galaxies. Among these massive stars, O-type stars represent the most extreme end of the stellar mass spectrum. These stars have effective temperatures that exceed 30,000 K and masses typically above 15 solar masses. They are characterised by intense ultraviolet radiation fields, powerful stellar winds, and short main-sequence lifetimes reaching only a few million years. Their spectra are characterised by the presence of ionised helium (He II) absorption lines, with early O-type stars showing strong He II and weak neutral helium (He I) features, while later O-type stars exhibit weaker He II and stronger He I lines. Although rare compared to lower-mass stars, O-type stars dominate the radiative and mechanical energy budget of star-forming galaxies and therefore have an effect that is disproportionate to their numbers (Langer 2012).

The importance of O-type stars extends across multiple areas of astrophysics. O-stars are unambiguous tracers of recent and ongoing star formation, since their short lifetimes mean they can often only be observed close to their birth sites within star-forming regions (Kennicutt 1998; Kennicutt & Evans 2012). Studies into O-stars therefore provide insight into star-forming regions and details such as the efficiency of gas collapse in molecular clouds (Zinnecker & Yorke 2007). O-stars occupy the upper end of the initial mass function (IMF), first described by Salpeter (1955), thus providing key constraints on the high-mass slope of the IMF.

The intense ultraviolet radiation produced by O-stars ionises surrounding hydrogen, generating H II regions that serve as observable signposts of massive star formation. A well-known example of this is the Orion Nebula (M42), the closest massive-star forming region to Earth. H II regions are often used as indicators for elemental abundances, however massive stars can also be used as an alternative to these regions to measure abundances (Simón-Díaz 2020). The ultraviolet output of O-stars dominates the hydrogen-ionising photon budget of young stellar populations, influencing nebular emission-line diagnostics and providing key constraints on the ionisation state of galaxies (Kennicutt & Evans 2012; Schaerer 2003). Additionally, through radiation pressure, photoionisation heating, and line-driven stellar winds, O-stars inject momentum and energy into the interstellar medium, regulating further star formation and contributing to the self-regulation of galactic discs (McKee & Ostriker 2007; Langer 2012). Due to their domination of this photon budget, quantifying the ionising photon output of massive stars is therefore essential for understanding the radiative feedback of stellar populations.

At the end of their lives, O-type stars undergo core-collapse supernovae, synthesising and dispersing heavy elements into the surrounding medium. In particular, they are the primary contributor to the production of α -elements such as oxygen, neon, and magnesium (Woosley & Weaver 1995). The enrichment of galaxies with oxygen is especially significant, as it underpins the chemistry of life as it is currently understood. In this way, O-stars are central not only to galactic evolution but also to the broader narrative of chemical evolution and the evolution of life. Blue supergiants, representing an evolved phase of massive stars, can also be used as standardised can-

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dles to measure distances via the flux-weighted gravity–luminosity relationship (Kudritzki et al. 2003), highlighting the importance of accurately determining their fundamental parameters.

The identification and study of O-type stars rely on both photometric and spectroscopic techniques. O-stars are intrinsically luminous and exhibit very blue colours as a result of their high effective temperatures. Photometric surveys therefore identify candidate massive stars through colour indices measured using observations with different filters, such as U–B and B–V, isolating the bluest and most luminous sources. Modern large-scale surveys, including missions such as Gaia, provide precise astrometry and photometry, enabling the construction of homogeneous stellar catalogues across nearby galaxies (Gaia Collaboration 2016). However, definitive classification requires spectroscopy. Spectral subtypes of massive stars are determined primarily from the relative strengths of He II and He I absorption lines, while luminosity classes are inferred from line widths and wind-sensitive features (Walborn & Fitzpatrick 1990; Simón-Díaz 2020).

In addition to optical studies, considerable effort has been devoted to observing massive stars in the far-ultraviolet, particularly with the Hubble Space Telescope, where key diagnostic spectral features are accessible. These wavelengths are especially important for interpreting observations of high-redshift galaxies, where rest-frame ultraviolet emission is redshifted into the optical and infrared regimes (Steidel et al. 2014).

From combined photometric and spectroscopic observations, a range of fundamental stellar parameters can be derived. Effective temperatures are typically calibrated against spectral subtype, while bolometric corrections allow the conversion of visual magnitudes into total stellar luminosities. However, such calibrations are not universal and depend sensitively on stellar abundances and metallicity. Metallicity influences line blanketing in stellar atmospheres and regulates mass-loss through radiatively driven stellar winds. Weaker stellar winds, associated with metal-poor environments, reduce angular momentum loss and allow massive stars to retain higher rotational velocities throughout the main-sequence phase (Meynet & Maeder 2000). Mass-loss rates for O-type stars have been observed to scale with metallicity, approximately as $\dot{M} \propto Z^{0.69 \pm 0.10}$ (Vink et al. 2001), with empirical estimates suggesting a relation closer to $\dot{M} \propto Z^{0.83 \pm 0.16}$ (Mokiem et al. 2007). This mass-loss can alter the evolutionary pathways of massive stars (Martins et al. 2005; Vink et al. 2001; Massey 2013). Consequently, stellar parameters derived for nearby massive stars in the Milky Way cannot necessarily be applied directly to environments with significantly different chemical compositions, making metallicity a key factor in the study of massive stars.

Mass-loss rates of stars are also influenced by stellar rotation, which can modify the distribution of radiative flux across the stellar surface and alter the structure of the stellar wind. Rotation also affects stellar evolution more broadly by modifying evolutionary tracks in the Hertzsprung–Russell diagram, lowering the effective gravity, extending the main-sequence lifetime, and enhancing the mixing of material produced through the CNO cycle to the stellar surface (Maeder 1997; Heger et al. 2000). Observational evidence for such mixing was found in the analysis of extreme O-type supergiants by Crowther et al. (2002), who reported strong nitrogen enrichment accompanied by depleted carbon and oxygen abundances, indicating the presence of CNO-processed material at the stellar surface. This rotational mixing is expected to be more efficient at lower metallicities (Georgy et al. 2013). These effects lead to further reduced line blanketing and consequently harder ultraviolet spectra, increasing

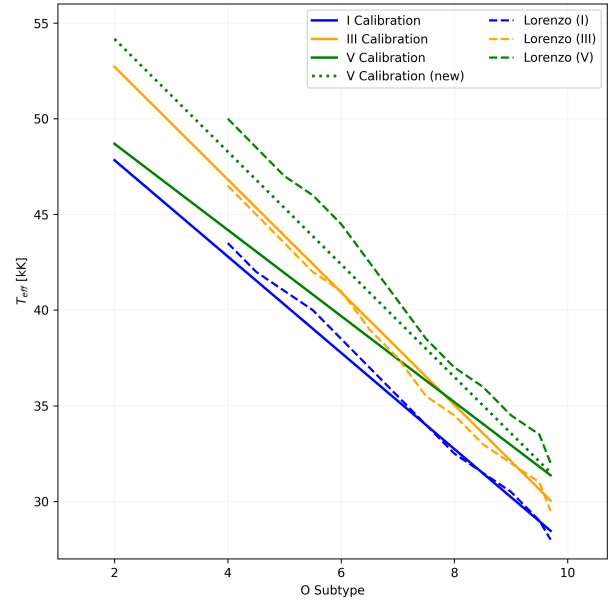


Figure 1. Effective temperature–spectral subtype calibrations for luminosity classes I, III, and V. Solid lines show the linear calibrations, while dashed lines indicate the corresponding relations from Lorenzo et al. (2025). The dotted green line shows the recalibrated dwarf (V) relation, anchored to late-type O dwarfs. An O Subtype of 10 is analogous to a star with spectral type B0.

the production of hydrogen-ionising photons per unit stellar mass (Schaerer 2003).

In this context, the Small Magellanic Cloud (SMC) provides an ideal laboratory for studying massive stars in low-metallicity environments. As a nearby dwarf irregular galaxy located at a distance of approximately 60 kpc (Graczyk et al. 2020), the SMC is sufficiently close that individual massive stars can be resolved and studied in detail. Its position outside the Galactic plane also reduces foreground extinction and line-of-sight confusion, making it easier to carry out large photometric and spectroscopic surveys. Crucially, the metallicity of the SMC is approximately one-fifth of the solar value (Russell & Dopita 1990), significantly lower than that of the Milky Way. In low-metallicity environments like this, the structure and evolution of massive stars are consequently altered through the mechanisms described above. Additionally, since galaxies at high redshift are typically metal-poor, studies into the effects of low-metallicity environments on stars provide important insight into the properties of massive stars in the early Universe (Steidel et al. 2014). As such, the SMC provides an important local analogue for studying stellar populations in low-metallicity environments, including those present in the early Universe.

Studies have also shown that a large fraction of massive stars are members of binary or higher-order multiple systems. Observational surveys of O-type stars indicate that the multiplicity fraction is extremely high, with estimates suggesting that 70% of O-stars are found in close binary systems (Sana et al. 2025). Given the young ages of O-type stars, this suggests that massive stars are commonly born in multiple systems. In many binary systems, the orbital separations are sufficiently small that binary interaction becomes inevitable during stellar evolution. Binary interaction therefore plays a major role in the evolution of massive stars, and can produce phenomena such as stellar mergers and mass gainers (de Mink et al. 2014). Studies indicate that approximately 20–30% of O-type stars will merge with

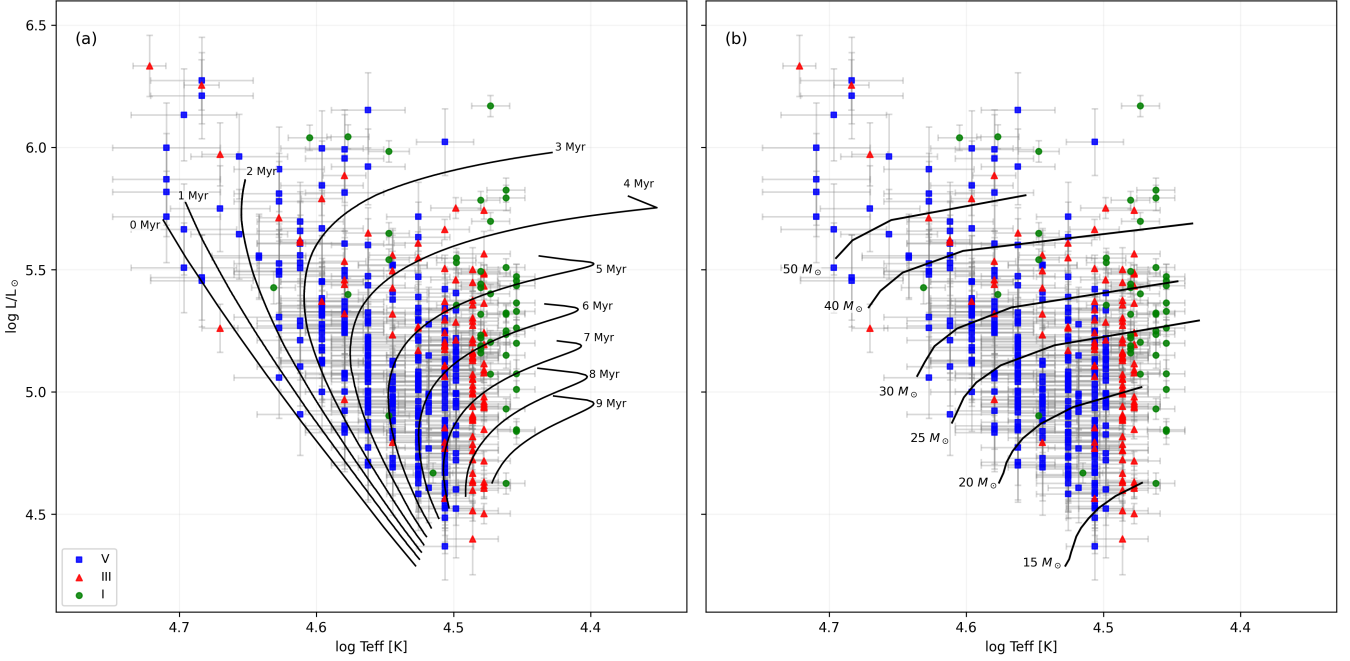


Figure 2. Hertzsprung-Russell Diagrams of O-type stars in the SMC. Panel (a) shows the stellar distribution with overlaid isochrones spanning 0–9 Myr, while panel (b) shows evolutionary tracks for selected initial masses. These are obtained and derived from [Brott et al. \(2011\)](#).

a companion during their lifetime, while 40–50% will either experience significant mass accretion or be stripped of their outer envelopes through binary interaction ([Sana et al. 2012](#)). Such processes can dramatically alter stellar evolution compared to the pathways predicted for single stars and may produce objects such as Wolf–Rayet stars ([Crowther 2007](#)). Consequently, the evolution of massive stars cannot be fully understood without considering the effects of binarity, which can influence both the physical properties of individual stars and the integrated feedback produced by massive stellar populations.

O-type stars within the SMC have been the subject of extensive surveys over several decades. Early photographic surveys by [Sanduleak \(1969\)](#) and [Azzopardi & Vigneanu \(1975\)](#) provided the first large-scale catalogues of massive stars in the Magellanic Clouds. These were later expanded through CCD-based surveys (e.g. [Massey et al. \(1989\)](#); [Massey \(2002\)](#)), which significantly improved photometric coverage and spectral classifications. More recent large-scale spectroscopic surveys, including VLT-FLAMES ([Evans et al. 2011](#)), RIOTS4 ([Lamb et al. 2016](#)), and BLOeM ([Shenar et al. 2024](#)), have provided increasingly homogeneous and detailed datasets for massive stars in the SMC. Nevertheless, significant challenges remain in constructing homogeneous datasets that combine spectral classifications, photometric measurements, metallicity-appropriate calibrations, and derived physical parameters. Temperature scales and bolometric corrections must be adapted to SMC metallicity, intrinsic colours may differ from Galactic standards, and uncertainties must be propagated consistently to produce reliable luminosities and ionising flux estimates.

A comprehensive understanding of O-type stars in the SMC therefore requires the integration of photometric catalogues, spectroscopic classifications, and metallicity-appropriate calibrations. By deriving effective temperatures and bolometric corrections, constructing Hertzsprung-Russell diagrams, and estimating hydrogen-ionising photon outputs, one can assess both the evolutionary distribution of

massive stars and their cumulative feedback into the surrounding medium. Such work contributes to improved models of massive star populations in low-metallicity galaxies and strengthens the connection between resolved stellar studies and integrated-light observations of distant systems.

This dissertation aims to investigate the physical characterisation of O-type stars in the Small Magellanic Cloud through the systematic combination of photometric and spectroscopic data, the application of low-metallicity calibrations, and the derivation of luminosities and ionising photon production rates. In doing so, it seeks to understand the massive star population of the SMC, clarify how metallicity influences the observable and physical properties of massive stars, and contribute to the broader understanding of stellar feedback and evolution in both nearby and high-redshift galaxies.

Section 2 describes the compilation and processing of the observational data used to construct the SMC O-star sample, and the derivation of low-metallicity calibrations and stellar parameters is presented in Section 3. In Section 4, these calibrations are applied to construct a Hertzsprung-Russell diagram for SMC O-type stars, and determine the hydrogen-ionising photon output for these stars. Section 5 presents the determination of stellar masses and ages using evolutionary models, along with a comparison between different model grids. Section 6 explores the spatial distribution of the O-star population within the SMC, including correlations with star-forming regions. Finally, Section 7 discusses the implications of these results in context with previous work, and Section 8 presents a brief summary of the main conclusions drawn from this work.

Table 1. Representative subset of the combined stellar parameter catalogue used in this work, showing both key stellar parameters and other determined parameters of interest.

ID	Spectral Type	T_{eff} (K)	$\log L/L_{\odot}$	$R/R_{\odot}$	Q_0 (s^{-1})	Mass ($M_{\odot}$)	Age (Myr)
AzV 2b	O9.5 I	28966	5.32	18.24	4.53×10^{48}	25.55	6.98
Sk 10	O6.5 III(f)	39466	5.79	16.82	2.32×10^{49}	47.51	3.47
2dFS 5010	O9.2 V	32972	5.28	13.33	4.85×10^{48}	23.54	6.60
2dFS 5012	O8 V((f))	36504	5.31	11.24	6.69×10^{48}	25.86	5.19
Sk 16	O9 I	30224	5.32	16.63	4.96×10^{48}	25.79	6.78
Sk 18	O6 I(f) + O7.5	37770	6.04	24.57	5.69×10^{49}	65.24	2.97
Lin 106	O7.5 V(n)	37976	5.82	18.67	2.43×10^{49}	43.16	3.60
Sk 22	O9.7 IIe	28463	5.33	19.02	4.41×10^{48}	25.51	7.03
2dFS 5022	O8 V	36504	5.17	9.61	4.90×10^{48}	23.89	5.43
Lin 121	O9 III:e	32106	5.19	12.68	3.32×10^{48}	23.44	7.06
AzV 43	O9 II	30224	5.23	15.08	4.08×10^{48}	23.73	7.34

2 DATA AND SAMPLE SELECTION

2.1 Photometric Data Sources

To construct a comprehensive sample of O-type stars in the SMC, a catalogue-based approach was adopted, combining spectroscopic classifications with multi-source photometric data. The initial sample of SMC O-type stars was adopted from the catalogue of [Bestenlehner et al. \(2025\)](#), which provides a comprehensive and up-to-date census of all known O-stars in the SMC. This catalogue contains a total of 448 spectroscopically classified O-type stars, including positional information and spectral classifications.

Photometric data for all stars in this catalogue were obtained primarily from two large-scale surveys: the catalogue of [Massey \(2002\)](#) and the Magellanic Clouds Photometric Survey of [Zaritsky et al. \(2002\)](#). These catalogues were selected due to their wide spatial coverage of the SMC and the availability of consistent broadband Johnson-Cousins photometry and uncertainties, making them well suited for constructing a uniform dataset of massive stars.

For stars located in the densely populated region of NGC 346, supplementary photometry was obtained from the catalogue of [Massey et al. \(1989\)](#), which offers improved coverage in crowded fields. For four sources located in N81, no Johnson-Cousins photometric data was available. For these cases, HST photometry from [Heydari-Malayeri et al. \(1999\)](#) was used to mimic *V*- and *B*-band photometry. This was achieved by using the F547M band to estimate the *V*-band magnitude, and averaging the values of the F410M and F467M bands to estimate the *B*-band magnitude.

In a small number of additional cases where no photometric data were available from the primary catalogues, values were obtained via the SIMBAD Astronomical Database ([Wenger et al. 2000](#)) using positional queries.

2.2 Cross-Matching and Dataset Validation

The initial source catalogue was cross-matched with the Massey and Zaritsky photometric catalogues using celestial coordinates. For each star, matches were identified based on proximity in right ascension and declination, and the angular separation between catalogues was computed using the great-circle distance.

This procedure resulted in photometric data being successfully obtained for 406 out of 448 stars in the Massey catalogue and 411 out of 448 stars in the Zaritsky catalogue. A small number of objects initially lacked photometric measurements in either catalogue but were subsequently recovered using supplementary sources as described above.

To assess the consistency of the photometric datasets, a comparison was performed for stars with measurements available in both the Massey and Zaritsky catalogues. The *V*-band magnitudes demonstrate strong agreement between the two datasets, lying close to a linear one-to-one relation, indicating consistent photometric zero-points and good overall agreement. This high recovery rate shows good overall consistency between catalogues, a small number of unmatched sources highlight the challenges of combining datasets with differing spatial resolution and completeness.

A small number of outliers were also identified, primarily in cases where magnitudes in the Zaritsky catalogue were systematically brighter than those reported by Massey. These discrepancies are likely attributable to differences in spatial resolution and crowding effects between the surveys.

2.3 Adopted Photometry

A systematic approach was adopted to determine the final photometric values for each star. For stars with data available from only a single source, the available values were adopted directly. For objects with measurements available in both catalogues, the *B* – *V* colour indices were calculated and then compared directly. Where the difference between catalogues was less than 0.1 mag, the values were considered consistent and were combined by taking the weighted mean based on their associated uncertainties.

For stars with larger discrepancies, individual inspection was performed, taking into account reported uncertainties and local environmental factors such as crowding. In such cases, preference was generally given to the values from the Massey catalogue, with some exceptions, due to its higher spatial resolution.

This approach balances statistical consistency with practical considerations, ensuring that reliable measurements are retained while minimising the impact of systematic offsets between catalogues.

Overall, this process results in a largely complete and internally consistent photometric dataset for the SMC O-type star population, providing a robust foundation for the derivation of stellar parameters in subsequent sections.

3 DETERMINATION OF STELLAR PARAMETERS

3.1 Spectral Types

For the majority of sources, well-defined spectral types were readily available in the original catalogue. Any anomalies were assigned realistic subtypes (e.g. an object listed as “mid O” was assigned the

subtype 7). For stars with classes in a range (e.g. O5-O7), a median value was assigned.

Any sources without an assigned luminosity class were automatically assigned as dwarfs (class V). This likely generates a source of systematic errors in some sources where this assignment was incorrect. For systems with multiple stellar components (e.g. binaries), the spectral subtype and luminosity class were assigned based on the brightest component, under the assumption that it dominates the observed flux.

One source, AzV 332, was excluded from the analysis due to its complex spectral classification ("WN3:ha + O + O + O"), indicating a multiple system with a Wolf-Rayet star and several other massive components. In this case, the O-type components are not individually classified by spectral subtype, and no single component can be assumed to dominate the observed flux. It was therefore not possible to assign a representative spectral subtype and luminosity class using the methodology described above. This object was consequently omitted from the analysis, resulting in a final sample of 447 O-type stars.

3.2 Spectral Type–Temperature Calibration

Effective temperatures were calculated for all stars based on their spectral subtypes and luminosity classes. Empirical calibrations were constructed using a compilation of literature calibrations for O-type stars derived using FASTWIND (Santolaya-Rey et al. 1997), CMFGEN (Hillier & Miller 1998), and PoWR (Sander et al. 2012) model atmosphere codes. Only SMC stellar data was used for these calibrations. The full list of sources and studies used for this process is shown in Table A1.

For luminosity classes I, III, and V, temperature values were compiled as a function of spectral subtype and fitted with linear relations. Due to the lack of available data in the literature for stars of luminosity classes II and IV, it was not possible to carry out similar calibrations for these classes. Instead, it was deemed necessary to use the class I calibrations for any stars with luminosity class II, and the class III calibrations for any stars of class IV. These calibrations were carried out and are shown in Figure 1. The calibrations along with all the data points used to carry out the calibration fits are shown in Figure A1.

During this process, it was found that the initial calibration for dwarf (Class V) stars exhibited a significantly shallower slope than those for giants and supergiants. This discrepancy is attributed to the limited number of early-type O dwarfs available in the SMC, which results in an underconstrained fit at the high-temperature end. This was confirmed as a discrepancy in the dwarf calibration by comparing the calibrations to those from the low-metallicity study Lorenzo et al. (2025) (also shown in Figure 1), with the giant and supergiant calibrations having similar slopes to the previous study. This issue was addressed by adopting a similar slope (same as obtained for the calibrations for luminosity class III), and using an anchor of late O-stars to position the slope. The values of all O9.7 stars in the sample were averaged and used to anchor the slope. This ensures physically consistent behaviour across luminosity classes while remaining aligned with expectations from previous low-metallicity studies.

Uncertainties for these fits were estimated from the scatter of the calibration data about the fitted subtype–temperature relations. For each luminosity class, the residuals between the observed literature temperatures and the fitted values were calculated. The root-mean-square (RMS) scatter was used to characterise the typical uncertainty in this for each subtype. Earlier O-type stars are expected to exhibit greater intrinsic scatter in their physical properties than later sub-

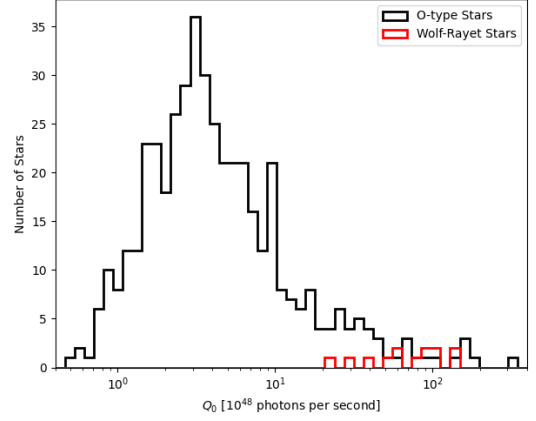


Figure 3. Histogram showing the total hydrogen ionising photon output for O-type stars in the SMC. Stars are distributed into logarithmic bins based on their value of Q_0 . The contribution from Wolf-Rayet stars is also shown.

types, reflecting the increased sensitivity of spectral diagnostics at high temperatures and the influence of stellar winds and multiplicity (Walborn et al. 2002). To account for this behaviour, as well as the overall scarcity of early-type stars compared to late-O stars in the sample, a subtype-dependent uncertainty model was adopted in which the uncertainties decrease smoothly from early- to late-type O stars. The uncertainty scale was anchored using the RMS scatter of the data at both the early- (O2-O4) and late-type (O8-O9.7) ends of the sample, with intermediate subtypes interpolated between these limits.

Due to a lack of any early-O supergiants in the SMC, there were no sources in the type I sample that could be used to calculate the RMS scatter in the early-type end of the sample. In this case, the global scatter of the dataset was used to anchor the uncertainty scale at the early end in order to avoid assigning unrealistically small uncertainties. This approach preserves the empirical spread of the literature data while enforcing physically reasonable behaviour across the full subtype range.

3.3 Bolometric Corrections

Bolometric corrections (BCs) were derived using the same method, based on literature values from the same studies. For each luminosity class, BCs were plotted against spectral subtype, with linear relations obtained for each class.

A similar issue was encountered with the dwarf calibrations, so the same manual adjustments were applied to ensure physically consistent behaviour. A similar slope to the other calibrations (same as class III) was adopted and a late-type anchor was used so that dwarf stars exhibited more negative bolometric corrections than giants and supergiants of the same spectral subtype as would be expected. This reflects the higher effective temperatures and correspondingly harder radiation fields of less evolved O-type stars, which result in enhanced ionising fluxes and more powerful radiation-driven stellar winds (e.g. Martins et al. 2005; Vink et al. 2001).

Uncertainties in the bolometric corrections were estimated from the scatter in the calibration datasets, using the same methodology as for the temperature calibrations.

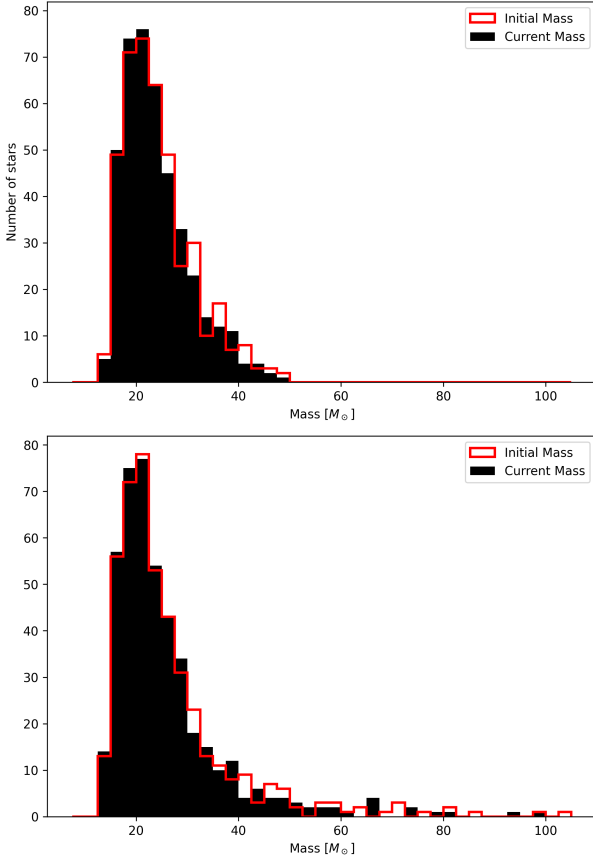


Figure 4. Mass distributions showing histograms comparing initial and current masses for the two BONNSAI models. The top panel shows data from the runs using the BONNSAI models of [Brott et al. \(2011\)](#); the bottom panel shows data from the runs using the [Hastings et al. \(2021\)](#) (MESA) models.

3.4 Extinction, Absolute Magnitudes, and Luminosities

Colour excesses were calculated for each star using observed and intrinsic colour indices, with intrinsic values assigned based on spectral subtype and luminosity class using low-metallicity calibrations from [Lorenzo et al. \(2025\)](#) which provides values for spectral types of O4 or later. The few stars with earlier spectral types than O4, were assigned the same intrinsic colour values as O4 stars of that luminosity class. Extinction in the V -band was derived assuming a standard extinction law, with a value of R_V of 2.74 ± 0.13 used for the SMC ([Gordon et al. 2003](#)).

Absolute magnitudes were calculated from these using an adopted SMC distance modulus of 18.977 ± 0.0322 from [Graczyk et al. \(2020\)](#). Bolometric magnitudes were obtained as standard using the corresponding bolometric corrections. These were then converted into stellar luminosities in solar units.

3.5 Uncertainty Estimation

Uncertainties on all derived quantities were estimated using a Monte Carlo approach in order to account for nonlinear transformations and physical constraints within the calculation pipeline.

For each star, 5000 realisations were generated by sampling all input parameters from Gaussian distributions centred on their measured values, with widths given by their uncertainties. These parameters include the apparent V -band magnitude, observed colour,

intrinsic colour, and bolometric correction. In addition, global parameters such as the adopted SMC distance modulus and extinction coefficient were also sampled in each realisation to account for their uncertainties.

In each realisation, the colour excess and extinction were recalculated. A physical constraint was imposed such that the colour excess could not take negative values, ensuring that extinction remained physically meaningful. All derived quantities, including absolute magnitude, bolometric magnitude, and luminosity, were recomputed for each realisation.

Final values were taken as the median of the resulting distributions, while uncertainties were estimated from the spread of the distributions using the 16th and 84th percentiles. This approach provides a more realistic estimate of uncertainties than analytic error propagation, particularly in cases where truncation effects and nonlinear relationships introduce asymmetry into the resulting distributions.

4 STELLAR PROPERTIES AND IONISING PHOTON OUTPUT

4.1 Hertzsprung–Russell Diagram

Using the properties derived in Section 3, the Hertzsprung–Russell diagram (HRD) for the SMC O-type star sample was constructed. The resulting distribution shows a well-defined sequence spanning the expected temperature and luminosity range of massive stars, with effective temperatures extending from late- to early-type O stars and luminosities covering several orders of magnitude.

There is a large proportion of stars that occupy the main-sequence region, consistent with the short lifetimes of O-type stars and the expectation that most stars are observed during this phase. A number of objects are also found at higher luminosities and cooler temperatures, reflecting the expected evolutionary progression of stars on the HRD. Many of these correspond to giant and supergiant stars which have evolved off the main-sequence entirely. These can be seen more clearly in Figure 2.

The separation of these luminosity classes across the HR diagram further supports the validity of the adopted spectral-type calibrations in this study, with supergiants occupying systematically higher luminosities than dwarfs at lower temperatures.

4.2 Comparison with Evolutionary Models

Theoretical isochrones and evolutionary tracks appropriate for SMC metallicity were overlaid on the HRD. Isochrones spanning ages of 0–9 Myr were obtained from non-rotating SMC metallicity grids from [Brott et al. \(2011\)](#). These grids were also used to define evolutionary tracks for stellar masses in the range $15\text{--}50 M_\odot$. The isochrones and evolutionary tracks can be seen in Figure 2.

The observed stellar distribution shows good agreement with the model predictions. Most stars lie below the $50 M_\odot$ evolutionary track, indicating that the sample is dominated by moderately massive O-type stars. A well-defined zero-age main sequence (ZAMS) can be seen, with only a few stars lying directly on this isochrone. The overall morphology of the HRD is consistent with a young stellar population, with the majority of stars located between isochrones corresponding to ages of a few Myr.

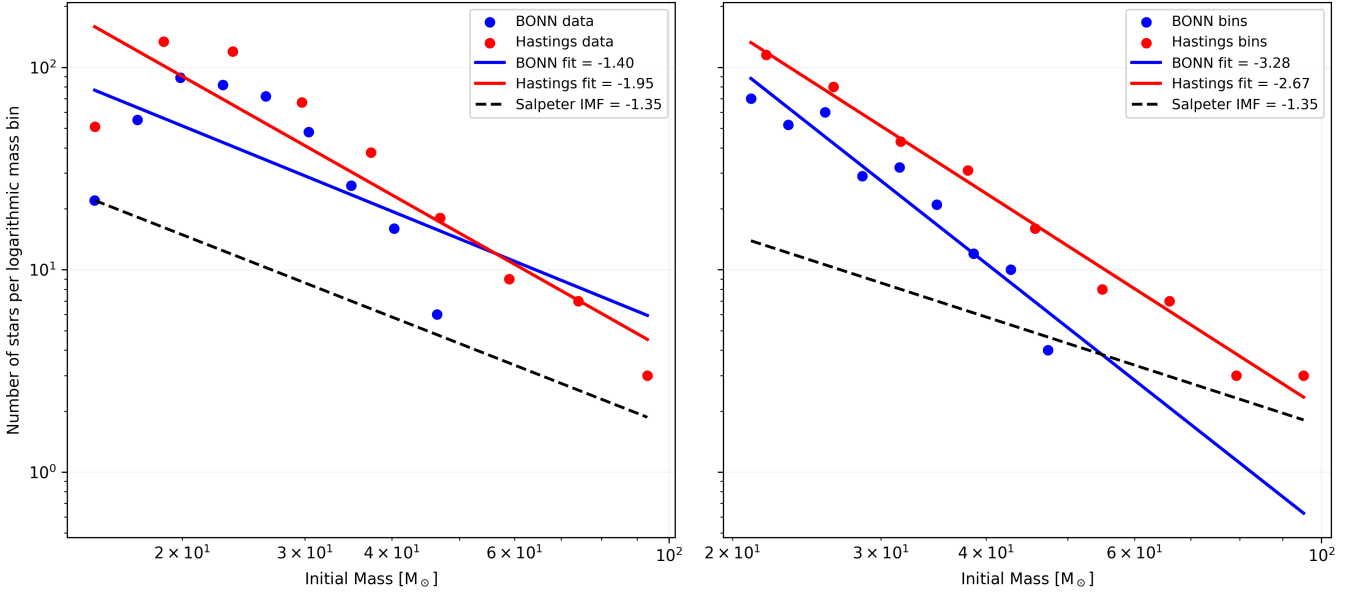


Figure 5. Initial mass functions derived from the two BONNSAI models, plotted in logarithmic space. The left panel shows the full mass range, while the right panel shows the high-mass regime ($M \geq 20 M_{\odot}$). Points indicate logarithmically binned star counts, and solid lines show the corresponding linear fits. The dashed line represents a Salpeter slope of -1.35 , normalised for visual comparison.

4.3 Population Characteristics

The distribution of stars across the HRD indicates a relatively continuous spread in both temperature and luminosity, reflecting a range of stellar masses and evolutionary states within the sample. While a minority of stars lie close to the ZAMS, most are offset towards slightly higher luminosities, suggesting that they have evolved away from their initial positions.

The lack of a significant concentration of stars directly on the ZAMS indicates that the population is not dominated by extremely young objects. Instead, the sample appears to be composed primarily of stars with ages of a few Myr, in agreement with expectations for massive star populations in active star-forming environments (e.g. Langer 2012; Massey 2013). The relative scarcity of stars located directly on the youngest isochrones may suggest either that the current SMC census is incomplete at the earliest evolutionary stages, or that the dominant episode of star formation in the SMC occurred some Myr ago.

A small number of evolved stars occupy positions on the HR diagram that are inconsistent with their assigned luminosity classes, such as a small number of objects classified as giants lying close to the zero-age main sequence. Possible causes include uncertainties in extinction and photometric measurements, which can lead to underestimates of stellar luminosity, or misclassification of luminosity class in spectroscopic catalogues. The effects of unresolved multiplicity may also contribute to inconsistencies between the inferred stellar parameters. These outliers therefore highlight the importance of classification accuracy in constructing physically consistent stellar populations.

In addition to these effects, the automatic assignment of luminosity class V to stars lacking a prior classification (Section 3.1) may introduce more subtle systematic biases in the stellar parameter estimates. In such cases, evolved stars incorrectly treated as dwarfs may be assigned inappropriate luminosities, potentially leading to less ob-

vious inconsistencies in the HR diagram, such as objects appearing at unusually high luminosities for their inferred evolutionary state.

4.4 Ionising Photon Output of SMC O-type Stars

The total ionising photon output of the SMC O-type star population was estimated using empirical relations between spectral subtype and hydrogen-ionising photon flux derived for low-metallicity massive stars by Lorenzo et al. (2025). These relations, defined separately for luminosity classes I, III, and V, provide the ionising photon production rate per unit surface area, q_0 , and are well suited to this study given the low-metallicity environment of the SMC.

As discussed in Section 3.2, the temperature calibrations shown in Figure 1 show good agreement in slope with those presented by Lorenzo et al. (2025), providing consistency between the adopted stellar parameters and the q_0 values. This is particularly important given the strong dependence of ionising flux on effective temperature. The total ionising photon output, Q_0 , was calculated using these values as shown in Equation 1. Stellar surface areas were derived from the stellar temperatures and luminosities using the Stefan–Boltzmann relation.

$$Q_0 = 4\pi R^2 q_0 \quad (1)$$

The resulting ionising photon rates span the expected range for O-type stars, with typical values of order 10^{48} – 10^{50} photons per second, and the distribution of Q_0 values is shown in Figure 3. This distribution shows a peak in the distribution at an approximate value of $Q_0 = 3 \times 10^{48} \text{ ph s}^{-1}$, consistent with the typical values for O-type stars. Summing over the full sample yields a total ionising photon output of approximately $(4.37 \pm 0.31) \times 10^{51} \text{ ph s}^{-1}$ from O-type stars alone. Uncertainties on Q_0 were estimated using standard error propagation, combining the uncertainties in stellar radius (derived from luminosity and effective temperature) with the uncertainties on the q_0 calibrations from Lorenzo et al. (2025).

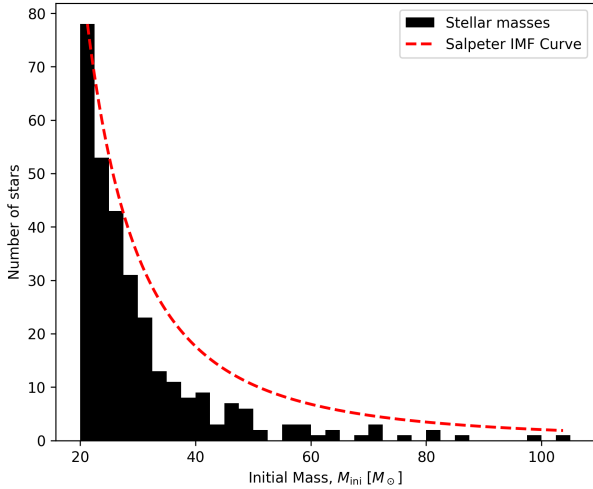


Figure 6. Histogram of the mass distribution derived from BONNSAI using the Hastings et al. (2021) evolutionary models, ranging from the peak of the distribution ($\sim 20 M_{\odot}$) to the high-mass end of the distribution. A normalised Salpeter (1955) initial mass function is also shown for comparison.

An additional contribution from Wolf–Rayet (WR) stars was estimated using published physical parameters for SMC WR stars (Hainich et al. 2015; Shenar et al. 2016). The corresponding ionising photon production rates were provided by the project supervisor as follows. In these models, the Lyman continuum photon output depends on both effective temperature and wind density, but is typically of order $\log Q_0 \approx 49.1$ for a hot WR star with $\log(L/L_{\odot}) \approx 5.3$. Scaling these values to the observed luminosities of individual SMC WR components yields a total contribution of approximately 10^{51} ph s^{-1} from these stars. The contribution of WR stars is also shown in Figure 3.

Including this contribution increases the total ionising photon output of the SMC massive star population to approximately 5.38×10^{51} ph s^{-1} , indicating that WR stars provide a significant, though secondary, contribution to the overall ionising budget.

These values of Q_0 obtained for individual stars are consistent with expectations for a young, massive stellar population. The total output obtained from this work can be compared to alternative estimates for the SMC based on the observed $H\alpha$ luminosity. $H\alpha$ emission arises from recombination in ionised hydrogen gas, and therefore can provide a measure of the rate at which Lyman continuum photons are produced by massive stars. Kennicutt et al. (2008) report a value of $\log L(H\alpha) = 39.67$ erg s^{-1} , which can be converted to an ionising photon production rate using $Q_0 = 7.4 \times 10^{11} L(H\alpha)$, yielding an approximate value of $Q_0 = 3.5 \times 10^{51}$ ph s^{-1} . This value is $\sim 35\%$ lower than the total ionising photon output obtained from the stellar population in this study.

This discrepancy likely reflects the fact that $H\alpha$ emission traces only the recombination of ionised gas, and may therefore underestimate the true ionising photon budget due to features such as photon leakage and dust absorption. Despite this, the results presented here highlight the dominant role of O-type stars and their evolved counterparts in shaping the ionising radiation field of the SMC. While this comparison provides a direct consistency check within the SMC, a broader comparison with other star-forming environments is presented in Section 7.

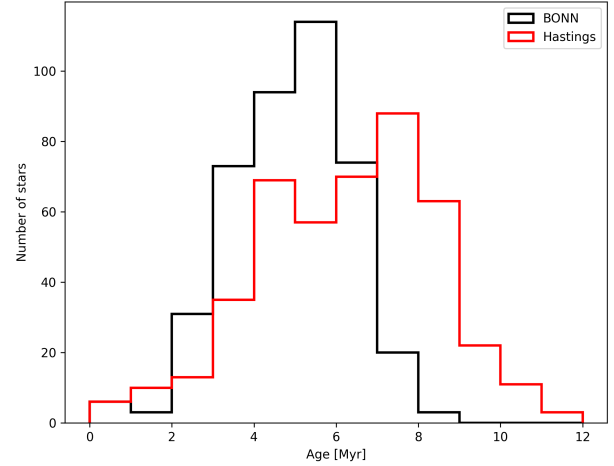


Figure 7. Histograms comparing the age distributions for the two BONNSAI models.

5 MASSES AND AGES

5.1 BONNSAI Modelling

Stellar masses and ages were estimated using the BONNSAI Bayesian inference tool (Schneider et al. 2014), which compares observed stellar parameters with predictions from stellar evolution models. The input parameters provided for each star were the effective temperature and luminosity, along with their associated uncertainties derived as described in Section 3.

An updated version of BONNSAI was used, adopting two sets of evolutionary models: the BONN-SMC grids of Brott et al. (2011), appropriate for low-metallicity environments, and MESA-based models presented by Hastings et al. (2021), who employed a setup similar to that of Schootemeijer et al. (2019). The BONN-SMC models are limited to initial masses below approximately $60 M_{\odot}$, resulting in a subset of the full sample being successfully modelled within this framework. The Hastings (MESA) models extend to higher masses, providing coverage for the most massive stars in the sample.

Overall, reliable mass and age estimates were obtained for all stars which had reliable temperature and luminosity data (447/448), allowing for a comprehensive analysis of the population properties.

5.2 Mass Distributions and Initial Mass Function

The resulting mass distributions were analysed for both the initial and current stellar masses. The population mass distributions for both models are shown in Figure 4. The two model sets show strong agreement, with differences in inferred masses typically at the level of $\sim 3\%$ (Appendix B). The majority of stars are found at relatively modest masses for O-type stars, with a significant fraction below $30 M_{\odot}$, and a clear decline in the number of stars toward higher masses.

To quantify the mass distribution, the slope of the IMF was estimated using the initial masses derived from the BONNSAI results. Nine bins were constructed in logarithmic mass space, and a linear fit was applied. This procedure was repeated over different mass ranges to assess the sensitivity of the inferred slope.

The resulting slopes are shown in Figure 5, alongside the Salpeter (1955) IMF for comparison. Over the full mass range, slopes of -1.95 and -1.40 were obtained for the Hastings and BONN-SMC models respectively, compared to the Salpeter value of -1.35 . When

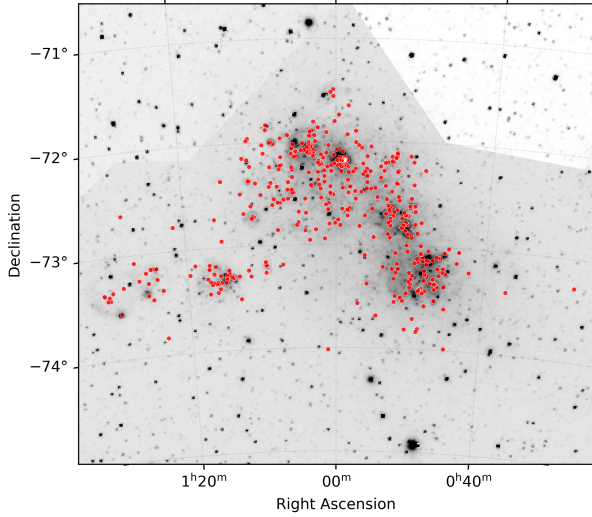


Figure 8. Spatial distribution of the SMC O-type star sample overlaid on an $H\alpha$ map of the SMC (Gaustad et al. 2001), highlighting $H\text{ II}$ regions of star formation.

a lower mass cutoff of $M > 20 M_{\odot}$ is applied, significantly steeper slopes of -2.67 and -3.28 are obtained for the Hastings and BONN-SMC models respectively, indicating a more rapid decline at higher masses.

Despite the strong agreement between the stellar masses derived from the two model grids, the Hastings models yield a steeper IMF over the full range. As these models extend to higher masses than the BONN-SMC grid, this steepening is attributed to the small number of stars present at the highest masses. The increase in slope when applying the low-mass cutoff further supports this interpretation, demonstrating that the inferred IMF is highly sensitive to the inclusion of a small number of high-mass objects.

The linear mass distribution for the Hastings models is shown alongside a Salpeter IMF for $M > 20 M_{\odot}$ in Figure 6. The Salpeter curve is normalised to the first bin for comparison. The observed distribution lies systematically below the Salpeter prediction at higher masses, further indicating a relative deficit of high-mass stars.

Overall, these results suggest that the apparent deviation from a Salpeter IMF is driven primarily by limited sampling of the high-mass regime. Additional factors such as sample incompleteness and unresolved binaries may also contribute and are discussed further in Section 7.2, but small-number statistics at the high-mass end are likely to dominate this observed behaviour.

5.3 Age Distribution

The age distributions of the sample are shown in Figure 7 for both model sets. It can be seen that the age distributions span approximately 0–10 Myr, with a clear concentration of stars at intermediate ages of a few Myr. The Hastings age distribution is broader and exhibits a more extended tail towards older ages compared to the BONN results. This is likely due to known systematic differences in the inferred ages between the models typically around 30% (Appendix B), much larger than those seen in the mass estimates.

These differences suggest that the inferred stellar ages are sensitive to the choice of evolutionary model, with the Hastings models permitting a wider range of age solutions. This broader age distribution may reflect differences in the treatment of stellar evolution at low metallicity, where uncertainties in mass loss and rotation can

lead to divergent evolutionary timescales. In addition, this difference may also contribute to variations in the inferred mass distributions, as stellar lifetimes and evolutionary pathways directly influence the mapping between observed properties and initial masses.

The relative lack of stars at very young ages is consistent with the distribution observed in the HR diagram, where few objects lie close to the ZAMS. Overall, the age distribution supports the interpretation of the sample as a young but not extremely recent star-forming population, which is a broadly expected result.

5.4 Model Comparison

A comparison between the BONN-SMC and Hastings model results shows excellent agreement in the derived stellar masses, with differences typically at the level of a few per cent. In contrast, the inferred ages show larger discrepancies, with differences of order 30 percent. Both comparisons are shown in Appendix B. This behaviour is consistent with known sensitivities in stellar evolution models, where age determinations are more strongly affected by model assumptions than mass estimates.

Despite these differences, both model sets produce consistent overall trends in the mass and age distributions, reinforcing the robustness of the inferred population properties.

6 SPATIAL DISTRIBUTION OF SMC O-TYPE STARS

6.1 Spatial Distribution

The spatial distribution of the SMC O-star population was examined by projecting their positions onto a map of the SMC. A $H\alpha$ image was adopted for this purpose, as $H\alpha$ emission traces ionised hydrogen gas produced by the intense ultraviolet radiation of young, massive stars, and therefore provides a direct probe of active star-forming regions. The $H\alpha$ image was obtained from the Southern H-Alpha Sky Survey Atlas (SHASSA; Gaustad et al. 2001). The resulting distribution of O-stars, shown in Figure 8, exhibits a strong concentration within these regions, particularly in known star-forming complexes such as NGC 346.

This spatial correspondence reflects the role of O-type stars as the dominant sources of ionising photons, and therefore being closely associated with regions of recent star formation. Given their short lifetimes, the majority of O-type stars are expected to remain near their birth sites, and the observed clustering places constraints on their dynamical evolution within the SMC, indicating relatively limited spatial diffusion.

However, the presence of a small number of isolated objects in regions with little or no detectable dust or $H\alpha$ emission suggests that additional processes may be at play for a subset of the population. These may include dynamical ejection from clusters or displacement following the supernova disruption of binary systems (e.g. Eldridge et al. 2011). Alternatively, they may trace regions where stellar feedback has already cleared the surrounding gas, removing observable signatures of recent star formation.

Distinguishing between these scenarios would require kinematic information, but their general presence highlights the complexity of massive star evolution and feedback within the SMC. The observed distribution therefore provides further evidence that the sample traces star formation across the SMC, while also highlighting the role of dynamical processes in shaping the spatial distribution of massive stars.

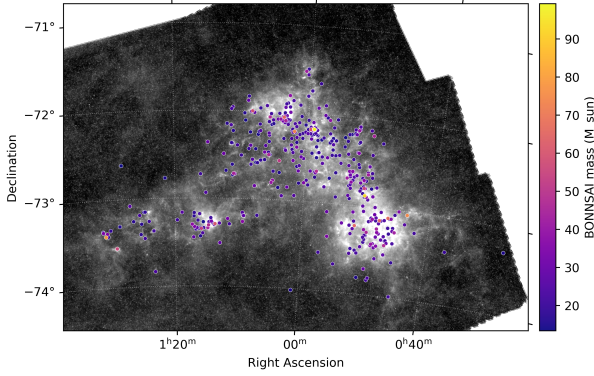


Figure 9. Spatial distribution of the SMC O-type star sample, overlaid on a Herschel SPIRE 350 μ m map of the SMC (Meixner et al. 2013). Points are colour-coded by current stellar mass, illustrating the variation of mass across the observed population.

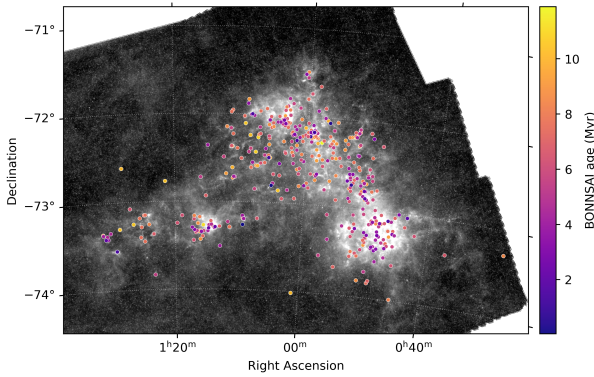


Figure 10. Spatial distribution of the SMC O-type star sample with points colour-coded by stellar age, overlaid on a Herschel SPIRE 350 μ m map of the SMC (Meixner et al. 2013).

6.2 Mass Distribution

The distribution was further examined as a function of stellar mass. As already known from the statistical distributions presented in Section 5, the majority of stars occupy the lower end of the O-type mass range, with relatively few objects present at the high-mass end of the spectrum.

No strong spatial segregation of highest-mass stars is observed, although the small number of such objects limits the statistical significance of any trends. In regions of high stellar density, particularly within clusters, the visibility of the most massive stars can be affected by projection effects. To aid visual interpretation, Figure 9 shows the spatial distribution with the most massive stars prioritised. For this, stars were overlaid on a Herschel SPIRE 350 μ m map of the SMC (Meixner et al. 2013) rather than the $H\alpha$ image used previously. This far-IR map clearly shows regions of cold dust in the SMC. Whilst cold dust may not be a direct tracer of ionised gas, it still traces the interstellar material which is associated with recent star formation, and therefore still provides useful environmental context for the positions of massive stars.

Overall, the spatial mass distribution is consistent with the steep high-mass decline inferred from the IMF analysis, with only a small number of very massive stars contributing to the upper end of the population. The absence of strong spatial segregation among the most massive stars suggests that, within the SMC, massive star formation

is not strongly concentrated, but is instead more spatially distributed. This is consistent with the irregular morphology of the SMC and its relatively modest star formation rate.

6.3 Age Distribution

The spatial distribution was also analysed as a function of stellar age, shown in Figure 10 overlaid over the same 350 μ m map of the SMC. The majority of stars are found to have ages of a few Myr, consistent with the age distributions in Section 5 and the positions of stars on the HR diagram. The relative lack of very young stars is again consistent with the absence of a strong concentration of objects near the zero-age main sequence on the HR diagram.

No clear large-scale age gradient is observed across the SMC, although localised variations are present within star-forming regions. This suggests that star formation has occurred in multiple regions across the galaxy in a spatially uncorrelated manner, rather than propagating in a single, coherent direction. This behaviour reflects the spatially complex structure of the SMC, where star formation is expected to be distributed rather than organised along well-defined structures such as spiral arms.

Overall, these spatial trends reinforce the interpretation that the SMC hosts a young, distributed massive star population shaped by a combination of local star formation processes, stellar feedback, and dynamical evolution, rather than a single dominant star-forming episode.

7 DISCUSSION

7.1 Overview of the SMC O-type Star Population

The results presented in this work provide a coherent picture of the O-type stellar population of the SMC. The Hertzsprung–Russell diagram, combined with the inferred masses, ages, and spatial distributions, indicates that the population is dominated by moderately massive stars with ages of a few Myr. The majority of stars are located slightly off the zero-age main sequence, suggesting that they have evolved away from their initial positions on the HR diagram but remain relatively young, as expected for O-type stars.

The strong spatial association of O-type stars with regions of $H\ II$, cold dust and active star formation further supports this interpretation. Together, these results indicate that the sample traces recent, but not extremely young, star formation across the SMC. Whilst the currently known sample of SMC O-type stars is likely close to complete, some objects may remain undetected. In particular, there is a possibility that very young massive stars may still be undetected within gas and dust, especially in active star-forming regions such as NGC 346. Although infrared observations have revealed ongoing star formation in these regions, the identification and spectroscopic confirmation of embedded O-type stars remains challenging, and such objects may therefore be underrepresented in current catalogues.

7.2 Implications of the Initial Mass Function

The derived IMF slopes are steeper than the Salpeter value, particularly when a low-mass cutoff is applied, indicating an apparent deficit of high-mass stars in the sample. While this could suggest a genuinely steeper IMF in the SMC, several effects are likely to cause this trend.

Firstly, the small number of very massive stars introduces significant statistical uncertainty at the high-mass end. In such regimes, the

inclusion or exclusion of only a few objects can significantly alter the inferred slope, making the results highly sensitive to these sampling effects.

In addition, unresolved binary and multiple systems are expected to contribute to biasing the slope. Massive stars in binary systems will have a high binary fraction, and unresolved systems like these may be interpreted as single stars with luminosities that do not correspond to a simple single-star mass. This can lead to systematic biases in mass estimates, either underestimating the true system mass or redistributing objects into incorrect mass bins. For example, it may preferentially reduce the number of intermediate-mass stars without significantly enhancing the high-mass population, contributing to a steepening of the inferred IMF.

It is also difficult to directly compare the IMFs produced by the two models, since the BONN-SMC grid is restricted to initial masses below $\sim 60 M_{\odot}$. This truncation can artificially bias the inferred IMF depending on how the high-mass tail is sampled, and complicates direct comparison with models that extend to higher masses.

Finally, observational incompleteness is likely to play a role. The SMC has a relatively modest and spatially distributed star formation rate, and the youngest, most massive stars may remain embedded deep in gas and dust and therefore undetected in optical surveys. This would further contribute to an apparent deficit of high-mass objects.

Together, these effects suggest that the observed steepening of the IMF is more likely to reflect a combination of statistical, observational, and model-dependent biases rather than a definitive deviation from a Salpeter-like IMF.

7.3 Ages, Star Formation History, and Feedback

The age distribution, spanning approximately 0–10 Myr with a concentration at a few Myr, is consistent with the form of the HRD and the lack of a strong population of stars located directly on the zero-age main sequence. This suggests that the observed population does not represent an extremely recent burst of star formation, but rather ongoing or slightly evolved star formation activity.

The absence of a clear large-scale age gradient across the SMC further indicates that star formation has occurred in multiple regions rather than propagating in a single direction. This behaviour is consistent with the irregular structure of the SMC, where star formation is expected to be spatially complex and influenced by past interactions with the LMC (Buckland-Willis et al. 2025; Mackey et al. 2017).

The dominant contribution of the ionising luminosity in the SMC arises from O-type stars, with Wolf–Rayet stars providing a considerable but secondary contribution (19%). The total ionising photon output derived for the O-type star population is consistent with expectations for a young, massive stellar population, and is comparable to the results of similar investigations. As discussed, the total ionising luminosity calculated here is found to be larger than previous estimates made from $H\alpha$ surveys, likely caused by $H\alpha$ measurements underestimating the true total photon output as a result of photon leakage and dust absorption.

In a more general context, comparisons with other regions can be made with studies that use similar methodologies to this study. In a census of the 30 Doradus star forming region of the LMC, Doran et al. (2013) find a total ionising photon output of 12.4×10^{51} ph s⁻¹ for 30 Doradus. This estimate is based on their full feedback-relevant sample of hot luminous stars, including additional objects beyond the O-type and Wolf–Rayet populations considered here, and is therefore not strictly like-for-like. The comparable number of O-type stars in both samples nevertheless provides a useful baseline

for comparison, indicating that differences in the ionising output are driven by stellar content and evolutionary state rather than simple population size. In particular, 30 Doradus contains a larger population of Wolf–Rayet and WN stars (31 compared to 13 in the SMC), which contribute approximately 40% of the total ionising photon output compared to 19% in the SMC. This enhanced population likely contributes significantly to the higher total flux observed in 30 Doradus (Larger than the SMC value derived in this work by a factor of approximately 2.3). This comparison also highlights the contrast between the more spatially distributed O-star population of the SMC, and the exceptional nature of 30 Doradus as a concentrated, high-intensity star-forming region. In both cases, the ionising output is sufficiently high to significantly influence the surrounding interstellar medium, supporting a picture in which massive star feedback plays a central role in regulating star formation.

In a similar investigation, Vacca et al. (1996) found a total ionising photon output of 7.0×10^{51} ph s⁻¹ for 429 O-stars in the Milky Way. A direct comparison is not straightforward, since this result is based on an incomplete sample of nearby O-stars rather than a coherent stellar population. Nevertheless, the comparable numbers of O-type stars, and the relatively similar magnitude of the ionising photon output may suggest a very broad consistency between the Milky Way and SMC O-star populations in this context.

These results highlight the key role of massive stars in shaping the interstellar medium through ionising radiation. As discussed in Section 1, ionising radiation and stellar winds from massive stars can disrupt surrounding gas in star-forming regions and inhibit further collapse (e.g. McKee & Ostriker (2007); Kennicutt & Evans (2012); Krumholz (2014)). Hence, the strong spatial correlation between O-type stars and regions of $H\alpha$ and cold dust appears to suggest that stellar feedback is actively influencing the surrounding environment, and may contribute to the regulation of subsequent star formation in the SMC.

The relative lack of very young stars discussed in Section 5 may reflect these effects. However, this remains uncertain since systematic issues such as observational incompleteness may also contribute to this trend. Very young massive stars may remain embedded within their natal gas and dust. Therefore these stars may be underrepresented in optical surveys, requiring IR surveys to be carried out in order to detect any newly born stars.

7.4 Limitations and Sources of Uncertainty

Several sources of uncertainty should be considered when interpreting the results of these studies. The assignment of spectral subtypes and luminosity classes introduces systematic uncertainties, particularly for stars with ambiguous classifications or unresolved multiplicity. The assumption that the brightest component dominates in binary systems may also bias the derived stellar parameters for systems with two or more stars of similar spectral types which contribute a similar share of the energy output of the system. The automatic assignment of luminosity class V to stars without a recorded classification introduces uncertainty for objects that in reality may belong to more evolved classes (I–IV). However, this approach was adopted as a pragmatic solution, motivated by the expectation that the majority of massive stars are observed during their main-sequence phase. Alternative assumptions would introduce similar uncertainties without clear justification, and this method therefore provided the most consistent and physically motivated baseline for the analysis.

The use of empirical calibrations for temperature and bolometric corrections introduces additional uncertainties, particularly at early spectral types where data are sparse. The subtype-dependent uncer-

tainty model discussed in Section 3.2 partially accounts for this, but residual systematic effects may remain. In addition systematic errors are immediately introduced for a subset of the stellar population due to calibrations only being performed for luminosity classes I, III, and V. While this was necessary due to limited numbers, it introduces unavoidable sources of uncertainty for stars of luminosity classes II and IV, where the type I and III calibrations were assigned.

The BONNSAI modelling is inherently dependent on the adopted stellar evolution models. While the comparison between the BONN-SMC and Hastings models provides some measure of robustness, the significant differences in inferred ages highlight the sensitivity of these results to model assumptions.

Finally, selection effects and observational incompleteness may influence the inferred distributions presented in this study, particularly at the high-mass end where the number of objects is small. Although extensive spectroscopic surveys of the SMC have been conducted in recent decades, such studies are inherently limited by observational constraints, including magnitude limits, crowding, and the requirement for high-quality spectra. As a result, a small number of O-type stars may remain undetected or unclassified. These factors should be considered when interpreting the IMF slope and the overall population properties.

8 CONCLUSIONS

This work has presented a large-scale census and analysis of the O-type stellar population of the SMC, combining multiple catalogues and calibrations to derive physical parameters for approximately 450 stars. Effective temperature and bolometric correction calibrations were constructed for multiple luminosity classes, allowing the determination of temperatures, luminosities, radii, ionising photon outputs, masses, and ages across the sample.

The resulting stellar population is found to be predominantly young, with ages generally below ~ 10 Myr. Comparisons between the BONN-SMC and Hastings evolutionary models show strong agreement in the derived stellar masses, while the inferred ages display larger systematic differences. The IMF slopes derived from both model grids are steeper than the canonical Salpeter slope, particularly when considering higher mass cutoffs. However, the analysis suggests that this behaviour is likely driven primarily by low-number statistics at the high-mass end, alongside additional effects such as unresolved binaries, observational incompleteness, and model-dependent limitations.

The spatial distributions show that the majority of O-type stars are concentrated near active star-forming regions and areas of enhanced dust emission, although a smaller number of apparently isolated massive stars are also identified. The derived ionising photon budget demonstrates that O-type stars dominate the radiative feedback of the SMC as expected, with Wolf–Rayet stars contributing a significant additional fraction despite their relatively small numbers.

Despite the large and comprehensive nature of the present sample, it is unlikely that the current census represents the complete population of O-type stars within the SMC. Existing surveys are expected to have identified the majority of massive stars, however some young and embedded objects are likely to remain undetected, particularly within dusty star-forming regions and crowded environments. Future surveys at infrared wavelengths will therefore be important for identifying obscured massive stars and obtaining a full, complete census of the SMC O-star population. Improved spectroscopic follow-up and higher spatial resolution observations will also help to better constrain unresolved multiplicity and reduce systematic uncertain-

ties in derived stellar parameters. Together, these observations will allow for a more complete analysis and understanding of the physical properties, evolution, and feedback of massive stars within the SMC.

Overall, the results of this study provide a consistent picture of a young, spatially distributed massive stellar population in the SMC. They highlight the role of the SMC as a key laboratory for understanding massive star populations in low-metallicity environments, with direct relevance to the interpretation of stellar populations in the early Universe.

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APPENDIX A: TEMPERATURE AND BOLOMETRIC CORRECTION CALIBRATION DATA

The temperature and bolometric correction calibrations were constructed using a compilation of literature studies of massive stars in the SMC. These studies utilise a range of non-LTE stellar atmosphere codes, including CMFGEN, FASTWIND, and PoWR, and provide empirical constraints on effective temperatures and bolometric corrections as a function of spectral subtype and luminosity class.

All stars used in the calibrations along with the temperature and bolometric correction data are listed in Table A1. Bolometric correction data were either readily available or manually calculated using the available magnitudes and luminosities presented in the studies.

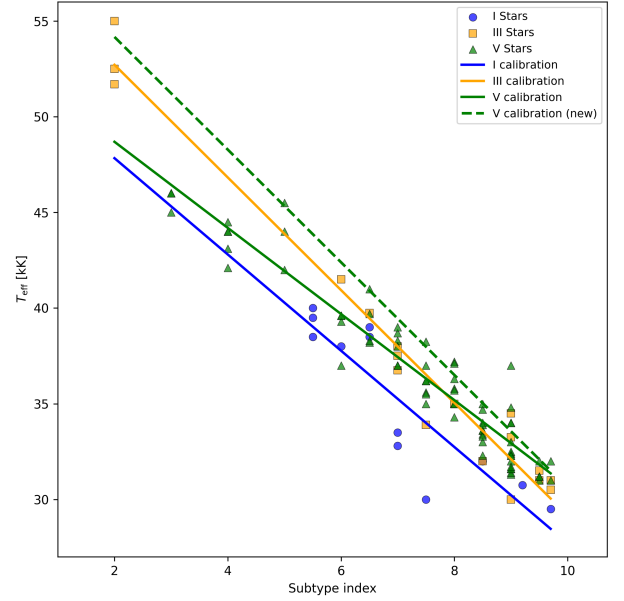


Figure A1. Effective temperature–spectral subtype calibrations for luminosity classes I, III, and V. Individual stars used for the calibrations are also shown. Solid lines show the linear calibrations, while the dotted green line shows the recalibrated dwarf (V) relation. The data for O9.7 dwarfs were used in order to anchor the slope for a more physically realistic fit.

IDs for the sources from the study by Rickard et al. (2024) were obtained from the Sabbi et al. (2007) catalogue which was used by the authors in this study. IDs from Ramachandran et al. (2019) were obtained from SIMBAD data for the objects following the links provided in the published catalogue.

An enhanced version of Figure 1 which includes all the data points used for the calibrations is shown in Figure A1.

Table A1: Literature values used to construct the effective temperature and bolometric correction calibrations. The studies from where the data were obtained, and the atmospheric code used to calculate them is also listed.

ID	Spectral Type	T_{eff} (K)	BC (mag)	Source	Atmospheric Code
AzV177	O4V((f))	44500	-4.14	Bouret et al. (2013)	CMFGEN
AzV 388	O4V	43100	-3.69	Bouret et al. (2013)	CMFGEN
MPG 324	O4V	42100	-3.92	Bouret et al. (2013)	CMFGEN
MPG 368	O6V	39300	-3.71	Bouret et al. (2013)	CMFGEN
MPG 113	OC6Vz	39600	-3.90	Bouret et al. (2013)	CMFGEN
AzV 243	O6V	39600	-3.80	Bouret et al. (2013)	CMFGEN
AzV446	O6.5V	39700	-3.80	Bouret et al. (2013)	CMFGEN
MPG 356	O6.5V	38200	-3.68	Bouret et al. (2013)	CMFGEN
AzV 429	O7V	38300	-3.45	Bouret et al. (2013)	CMFGEN
MPG 523	O7Vz	38700	-3.65	Bouret et al. (2013)	CMFGEN
MPG 046	O7Vn	39000	-3.62	Bouret et al. (2013)	CMFGEN
MPG 031	O8Vz	37200	-3.40	Bouret et al. (2013)	CMFGEN
AzV 461	O8V	37100	-3.25	Bouret et al. (2013)	CMFGEN
MPG 299	O8Vn	36300	-3.24	Bouret et al. (2013)	CMFGEN
MPG 487	O8V	35800	-3.37	Bouret et al. (2013)	CMFGEN
AzV 267	O8V	35700	-3.23	Bouret et al. (2013)	CMFGEN
AzV 468	O8.5V	34700	-3.17	Bouret et al. (2013)	CMFGEN
AzV 148	O8.5V	32300	-2.23	Bouret et al. (2013)	CMFGEN
MPG 682	O9V	34800	-3.21	Bouret et al. (2013)	CMFGEN
AzV 326	O9V	32400	-2.04	Bouret et al. (2013)	CMFGEN
AzV 189	O9V	32300	-2.53	Bouret et al. (2013)	CMFGEN
AV 14	O5 V	44000	-4.04	Massey et al. (2004)	FASTWIND
AV 207	O7.5 V((f))	37000	-3.53	Massey et al. (2004)	FASTWIND
AV 296	O7.5 V((f))	35000	-3.36	Massey et al. (2004)	FASTWIND
AV 377	O5 V((f))	45500	-4.14	Massey et al. (2004)	FASTWIND
AV 177	O4 V((f))	44000	-4.04	Massey et al. (2005)	FASTWIND
AV 435	O3 V((f))	45000	-4.12	Massey et al. (2005)	FASTWIND
AV 440	O8 V	37000	-3.52	Massey et al. (2005)	FASTWIND
AV 446	O6.5 V	41000	-3.82	Massey et al. (2005)	FASTWIND
MPG 324	O5 V	42000	-3.91	Massey et al. (2009)	FASTWIND
AV 435	O3 V((f*))	46000	–	Rivero González et al. (2012)	FASTWIND
AV 177	O4 V((f))	44000	–	Rivero González et al. (2012)	FASTWIND
AV 95	O7.5 V((f))	38250	-3.51	Bucks et al. (2024)	FASTWIND
AV 207	O7 V((f))z	38000	-3.58	Bucks et al. (2024)	FASTWIND
2dFS 3620	O9 V	32000	-3.29	Ramachandran et al. (2019)	PoWR
2dFS 3664	O9 V	33000	-3.29	Ramachandran et al. (2019)	PoWR
SMCSGS-FS 89	O9.5 V	31000	-2.26	Ramachandran et al. (2019)	PoWR
SMCSGS-FS 90	O9.7 V	31000	-2.91	Ramachandran et al. (2019)	PoWR
2dFS 3700	O8.5 V	35000	-3.38	Ramachandran et al. (2019)	PoWR
2dFS 3713	O9.7 V	32000	-3.13	Ramachandran et al. (2019)	PoWR
2dFS 3720	O9 V	34000	-3.86	Ramachandran et al. (2019)	PoWR
SMCSGS-FS 143	O8 V	35000	-3.44	Ramachandran et al. (2019)	PoWR
SK 183	O3 V((f*))z	46000	-4.18	Ramachandran et al. (2019)	PoWR
SMCSGS-FS 238	O8.5 V	34000	-2.86	Ramachandran et al. (2019)	PoWR
SMCSGS-FS 239	O9 V	34000	-3.69	Ramachandran et al. (2019)	PoWR
SMCSGS-FS 241	O7 V	37000	-3.21	Ramachandran et al. (2019)	PoWR
2dFS 3934	O8 Vz	35000	-3.36	Ramachandran et al. (2019)	PoWR
2dFS 3954	O6 V((f))z	37000	-3.54	Ramachandran et al. (2019)	PoWR
SMCSGS-FS 292	O7 V((f))z	37000	-3.33	Ramachandran et al. (2019)	PoWR
SMCSGS-FS 298	O9.5 V	32000	-3.38	Ramachandran et al. (2019)	PoWR
MPG 487	O7.5 V	35600	–	Rickard et al. (2024)	PoWR
MPG 476	O6.5 V	38300	–	Rickard et al. (2024)	PoWR
MPG 417	O7.5 V	36200	–	Rickard et al. (2024)	PoWR
MPG 471	O8.5 V	33000	–	Rickard et al. (2024)	PoWR
MPG 467	O7.5 V	35500	–	Rickard et al. (2024)	PoWR
MPG 370	O8.5 V	33600	–	Rickard et al. (2024)	PoWR
MPG 495	O8.5 V	33600	–	Rickard et al. (2024)	PoWR
MPG 451	O9.0 V	31300	–	Rickard et al. (2024)	PoWR
MPG 481	O8.5 V	33400	–	Rickard et al. (2024)	PoWR
MPG 500	O7.5 V	36200	–	Rickard et al. (2024)	PoWR
MPG 439	O9.0 V	31400	–	Rickard et al. (2024)	PoWR
MPG 455	O9.0 V	32500	–	Rickard et al. (2024)	PoWR

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Table A1 – continued from previous page

ID	Spectral Type	T_{eff} (K)	BC (mag)	Source	Code
MPG 375	O8.5 V	33900	–	Rickard et al. (2024)	PoWR
MPG 468	O8 V	34300	–	Rickard et al. (2024)	PoWR
MPG 429	O9.0 V	32300	–	Rickard et al. (2024)	PoWR
MPG 427	O9.5 V	31200	–	Rickard et al. (2024)	PoWR
MPG 443	O9.0 V	31600	–	Rickard et al. (2024)	PoWR
MPG 486	O9.0 V	31600	–	Rickard et al. (2024)	PoWR
MPG 493	O9.0 V	31700	–	Rickard et al. (2024)	PoWR
MPG 499	O8.5 V	33300	–	Rickard et al. (2024)	PoWR
MPG 477	O9.5 V	31200	–	Rickard et al. (2024)	PoWR
NGC 346–3	ON2 III(f*)	52500	-4.54	Walborn et al. (2004)	CMFGEN
MPG 355	ON2 III(f*)	51700	-4.66	Bouret et al. (2013)	CMFGEN
AV 77	O7 III	37500	-3.27	Bouret et al. (2021)	CMFGEN
AV 95	O7 III((f))	38000	-3.64	Bouret et al. (2021)	CMFGEN
AV 69	OC7.5 III((f))	33900	-3.30	Bouret et al. (2021)	CMFGEN
AV 47	O8 III((f))	35000	-3.23	Bouret et al. (2021)	CMFGEN
AV 307	O9 III	30000	-2.87	Bouret et al. (2021)	CMFGEN
AV 439	O9.5 III	31000	-3.71	Bouret et al. (2021)	CMFGEN
AV 170	O9.7 III	30500	-3.00	Bouret et al. (2021)	CMFGEN
AV 378	O9.5 III	31500	-3.06	Massey et al. (2004)	FASTWIND
AV 80	O6 IIIIn(f)p	41500	-4.04	Bucks et al. (2024)	FASTWIND
AV 15	O6.5 III(f)	39750	-3.76	Bucks et al. (2024)	FASTWIND
AV 69	OC7 III	36750	-3.50	Bucks et al. (2024)	FASTWIND
SK 187	O8.5 III	32000	-3.06	Ramachandran et al. (2019)	PoWR
SMCSGS-FS 315	O9.7 (III)e	31000	-3.36	Ramachandran et al. (2019)	PoWR
AV 75	O5.5 I(f)	38500	-3.46	Bouret et al. (2021)	CMFGEN
AV 15	O6.5 I(f)	39000	-3.73	Bouret et al. (2021)	CMFGEN
AV 232	O7 Iaf	33500	-3.16	Bouret et al. (2021)	CMFGEN
AV 83	O7 Iaf	32800	-3.10	Bouret et al. (2021)	CMFGEN
AV 26	O6 I(f)	38000	-3.63	Massey et al. (2004)	FASTWIND
AV 75	O5.5 I(f)	40000	-3.79	Massey et al. (2004)	FASTWIND
AV 469	O8.5 I(f)	32000	-3.12	Massey et al. (2004)	FASTWIND
AzV 15	O6.5 I(f)	38500	-3.66	Massey et al. (2009)	FASTWIND
AzV 75	O5.5 I(f)	39500	-3.75	Massey et al. (2009)	FASTWIND
AV 372	O9.2 Iab	30750	-2.96	Bucks et al. (2024)	FASTWIND
AV 327	O9.7 Ib	29500	-2.73	Bucks et al. (2024)	FASTWIND
SK 190	O7.5 In(f)p	30000	-3.13	Ramachandran et al. (2019)	PoWR

APPENDIX B: MASS AND AGE COMPARISON DIAGNOSTICS

This appendix presents a direct comparison of stellar masses and ages derived using the BONN-SMC and MESA evolutionary model grids for the same set of stars. These comparisons are included to assess the systematic differences between the two model frameworks. As discussed in the main text, the derived stellar masses show good agreement between the models, while the inferred ages exhibit larger discrepancies. The plots presented here in Figure B1 provide a visual representation of these trends. Plots were produced by Vincent Bronner.

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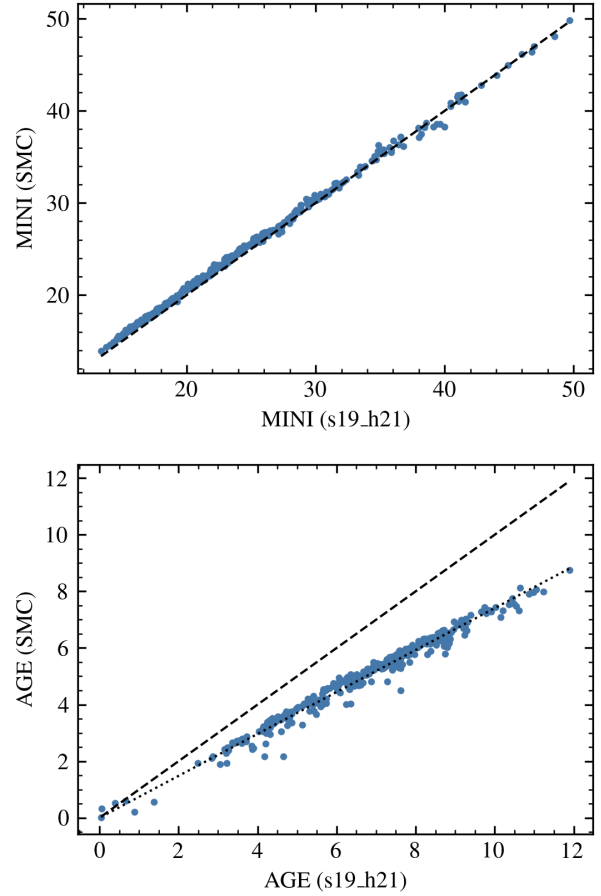


Figure B1. Comparison of stellar parameters derived from the BONN (y-axis) and Hastings (x-axis) models. The top panel shows comparisons of the initial masses of stars whilst the bottom panel shows comparisons of the ages between the models. Differences are shown for stars that are present in both datasets, highlighting systematic offsets between the two model grids.